EE 152-BASIC ELECTRONICS

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CHAPTER TWO

PN – JUNCTION DIODE

What to expect?

PN Junction → Biasing → Equivalent Circuit of a diode



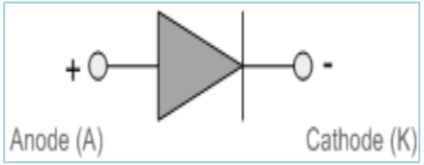
PN JUNCTION

A p-n junction is the junction between an n-type semiconductor and a p-type semiconductor

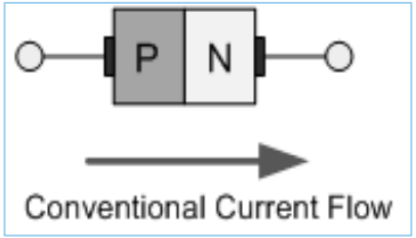


PN Junction

- A p-n junction is the junction between an n-type semiconductor and a p-type semiconductor.
- It is fabricated by introducing donor impurities into one side of an intrinsic semiconductor crystal and acceptor impurities into the other side.
- It is also called a semiconductor diode.

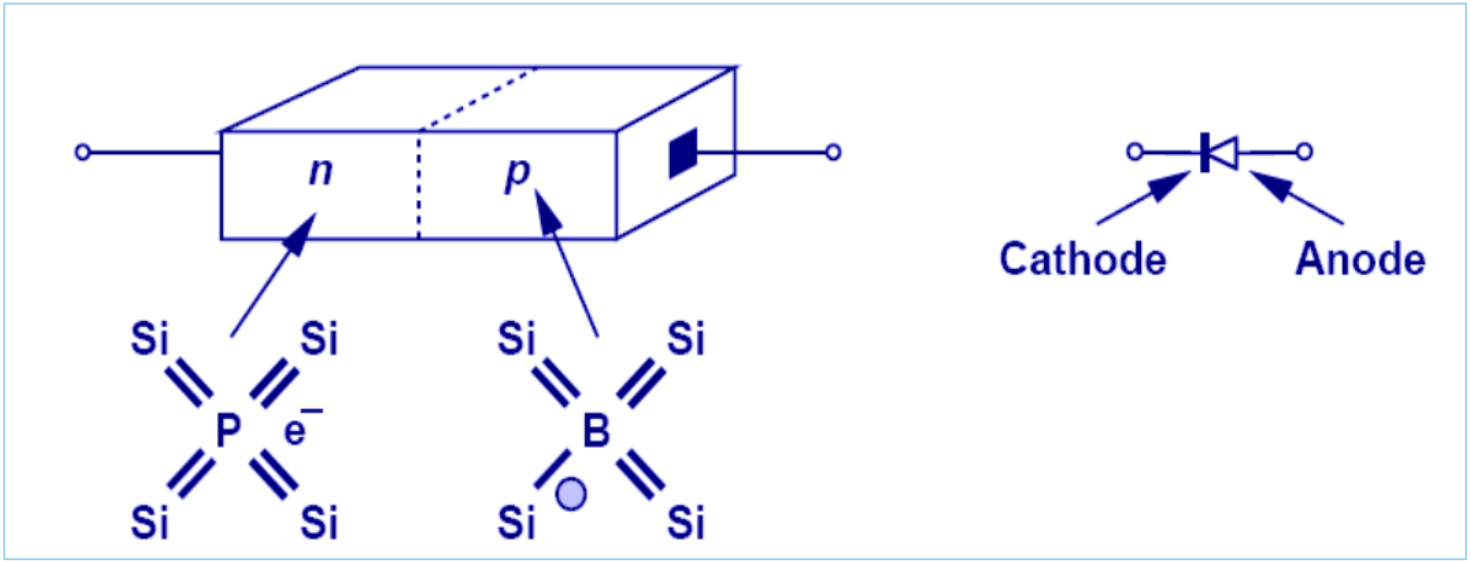


(a) schematic diagram



(b) block diagram

PN Junction

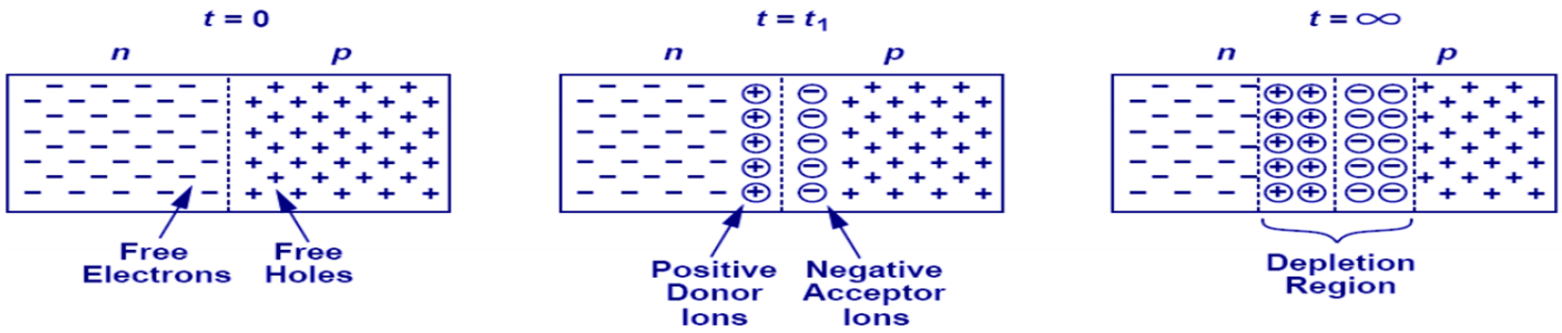


(a) When N-type and P-type dopants are introduced side-by-side in a semiconductor, a PN junction or a diode is formed.

(b) The positive terminal of the diode is known as the anode; the negative terminal is the cathode.

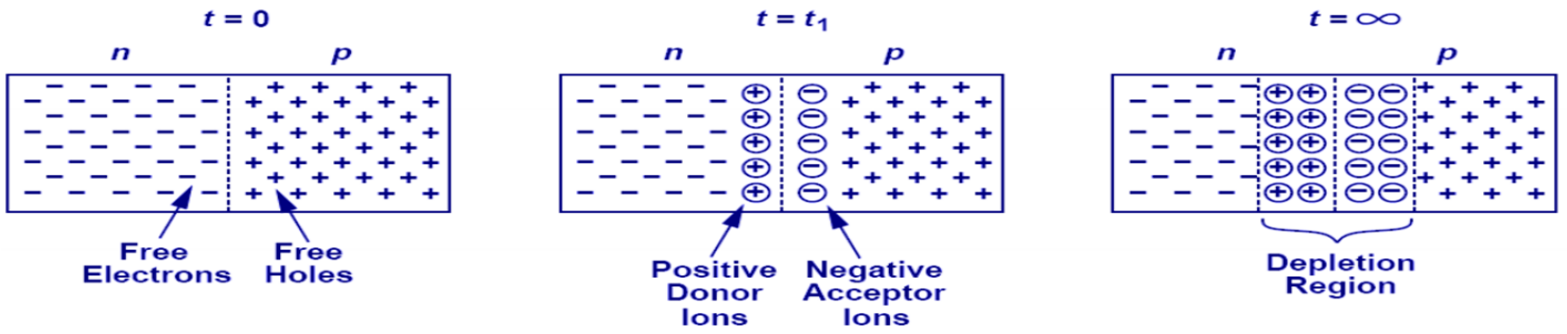
PN Junction

- The figure below illustrates the cross-section of a p-n junction.
- Assume the p-n junction is initially made (t = 0); A diffusion current consisting of both holes and free electrons will flow across the junction.
- Holes diffuse out of the p-type side and into the n-type side and free electrons diffuse out of the n-type side and into the p-type-side.



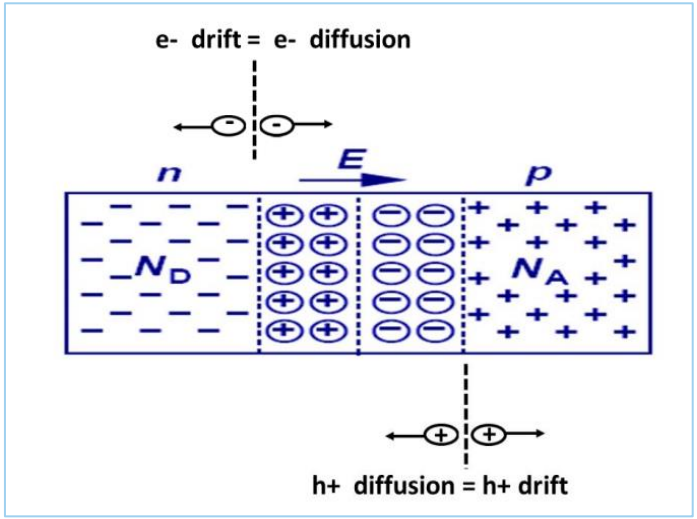
PN Junction

- The p-side becomes negatively charged and the n-side becomes positively charged ($t = t_1$), caused by the ionized acceptor atoms and ionized donor atoms on the p-side and n-side respectively.
- The charges on either side of the p-n junction cause an electric field to build up across the junction which is directed from the n-type side to the p-type side.



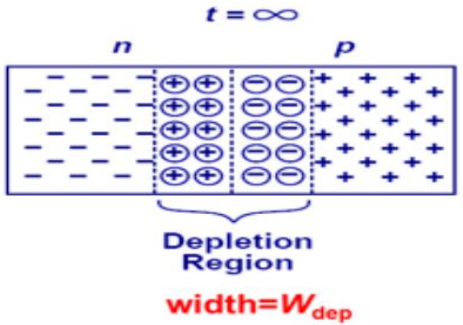
PN Junction in Equilibrium

- **The electric field opposes the diffusion of holes and electrons across the junction.**
- **Equilibrium is reached when the force exerted on the charge carriers by the electric field is equal to the diffusion force.**



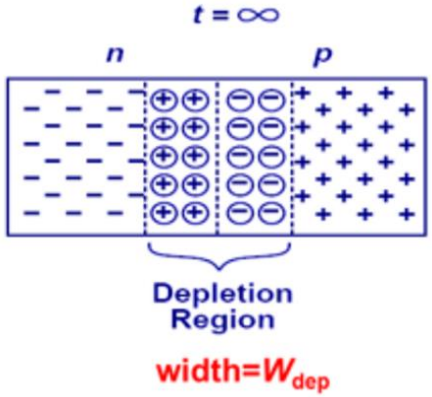
Depletion Region

- Other names are the space-charge region and the transition region.
- The width of the depletion region is called the depletion width (W_{dep}).



Junction Capacitance

- **Junction capacitance is the capacitance associated with the depletion region.**
- **The positive ions on the n-type side and the negative ions on the p-type side serve as the positive and negative plates respectively as in a parallel plate capacitor.**
- **The dielectric of the junction capacitance result from the depletion region space between the positive and negative ions.**



Junction Capacitance

- As in a parallel plate capacitor, the junction capacitance is given by:

$$C_{dep} = \frac{\varepsilon_o \varepsilon_r A}{W_{dep}}$$

Where

$\varepsilon_o\varepsilon_r = \varepsilon$ is the absolute permittivity of the semiconductor

A is the cross-sectional area of the junction and

W_{dep} is the depletion width

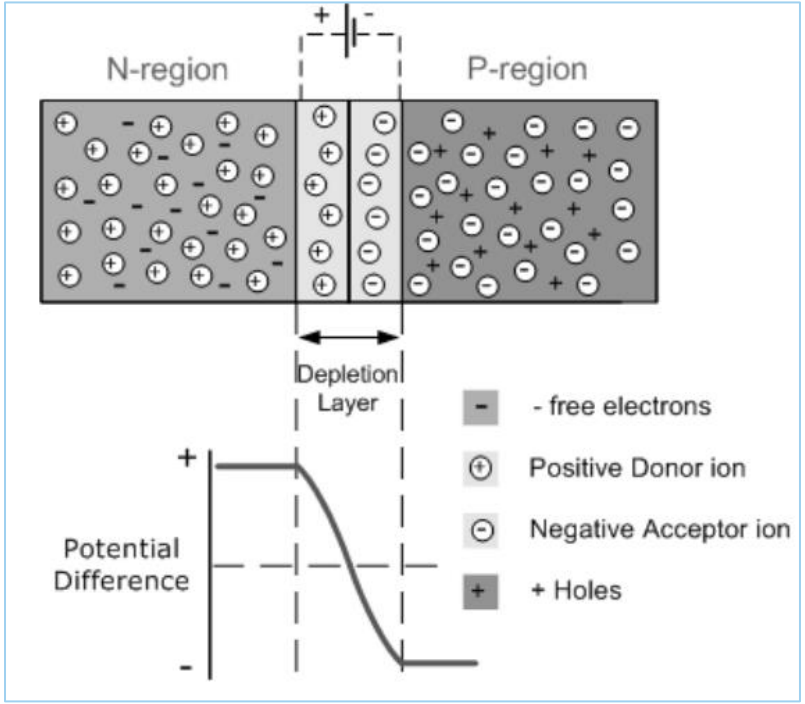
Built-In Potential

- Because there is an electric field in the depletion region of a p-n junction that is directed from the n-type side to the p-type side, it follows that there is a difference in potential or voltage across the junction.
- This voltage difference is called the built-in potential or contact potential or potential barrier.
- The built-in voltage V_B is given by:

$$V_B = V_T \ln\left(\frac{N_A N_D}{n_i^2}\right)$$

Where V_T is the thermal voltage.

Junction Capacitance



2. BIASING

Biasing establishes a voltage across the p-n junction diode terminals

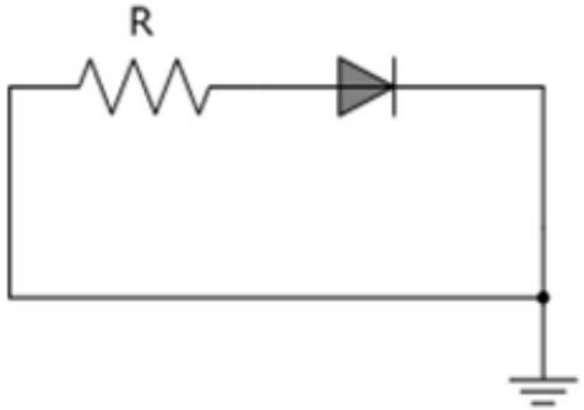
Biasing

- For semiconductors to conduct, a voltage potential is connected across it in a process called biasing.
- The external voltage, based on the how the biasing is done, can increase or decrease the potential barrier.
- There are three possible biasing conditions for the standard junction diode namely:

1. Zero biasing
2. Forward biasing
3. Reverse biasing.

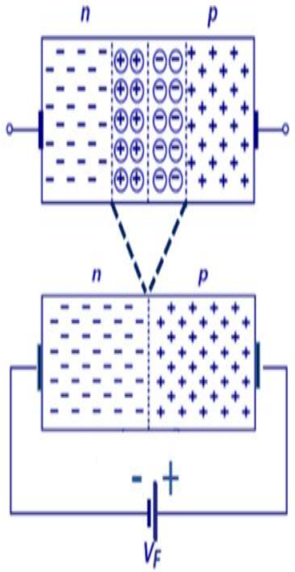
Zero Bias

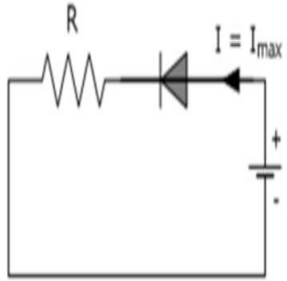
- **In zero bias condition, no external voltage is applied to the PN-junction.**
- **No current flows through the diode.**



Forward Biasing

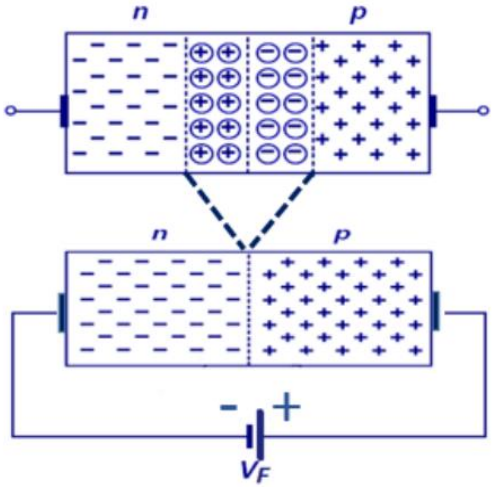
- When a diode is connected in a forward bias condition, a negative voltage is applied to the n-type material and a positive voltage is applied to the p-type material.
- If this external voltage becomes greater than the value of the potential barrier, 0.7V for Silicon and 0.3 V for Germanium, the potential barrier's opposition will be "subdued" and current will start to flow.





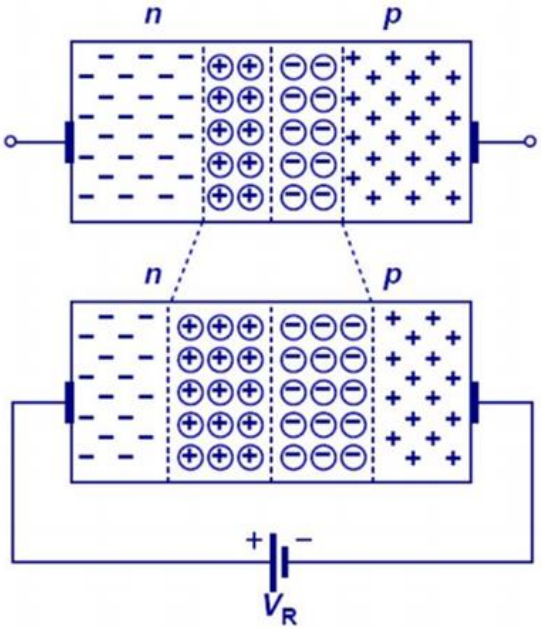
Forward Biasing

- The negative voltage repels electrons towards the junction giving them the energy to cross over and the holes are repelled in the opposite direction towards the junction by the positive voltage.
- These result in the depletion layer becoming very thin and narrow representing a low impedance path. In that event, a very small potential barrier is produced, allowing high currents to flow.
- The voltage at which the diode starts conducting is called the knee voltage or cut-in voltage

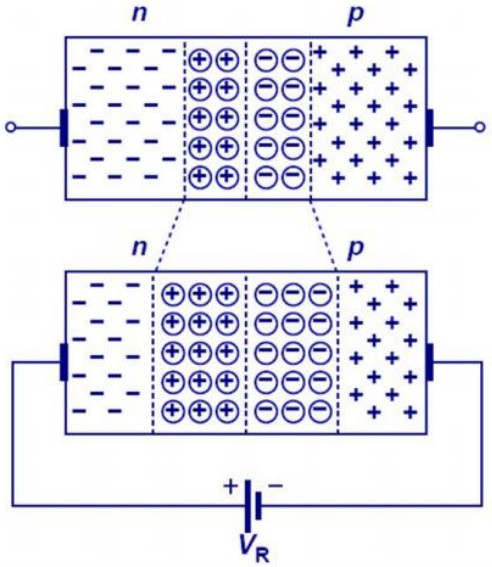


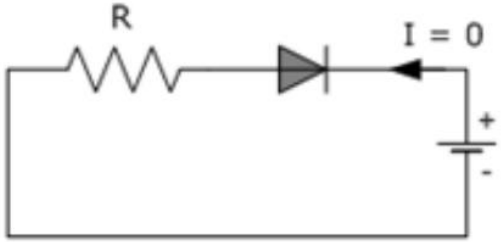
Reverse Biasing

- To reverse bias a diode, a positive voltage is applied to the N-type material and a negative voltage is applied to the P-type material as shown on the right.
- The positive voltage applied to the N-type material attracts electrons towards the positive electrode and away from the junction, while the holes in the P-type end are also attracted away from the junction towards the negative electrode.
- The net result is that, the depletion layer grows wider due to a lack of electrons and holes and presents a high impedance path.



Reverse Biasing



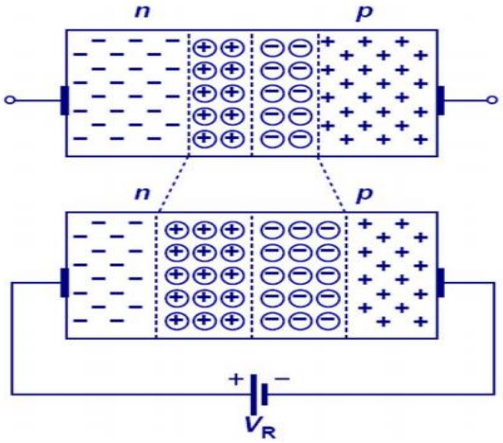


- **Consequently, a high potential barrier is created thus preventing current from flowing through the semiconductor material.**

Reverse Biasing

- However, a very small leakage current does flow through the junction that can be measured in microamperes, (μ A).
- This leakage current is due to drift of thermally produced electron-hole pairs across the junction i.e. the electric field across the junction causes the electron to be swept to the n-type side and the hole to be swept to the p-type side.
- If the reverse bias voltage, V_R applied to the junction is increased to a sufficiently high enough value called the Reverse Breakdown Voltage, it will cause the PN-junction to overheat and fail.

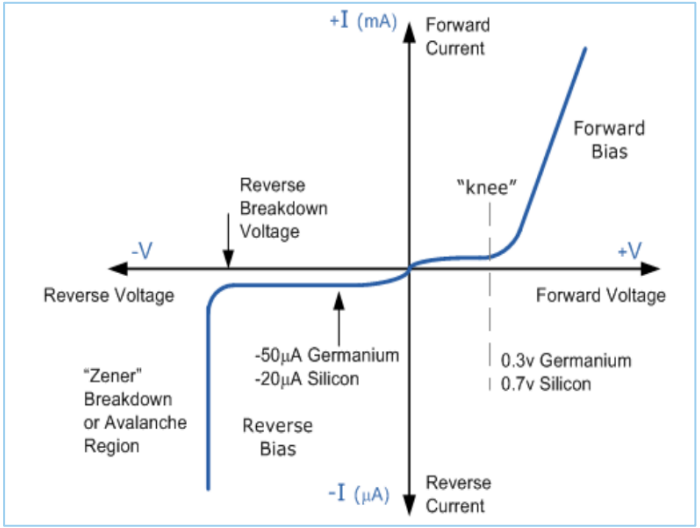
Reverse Biasing



- **This may cause the diode to become shorted and will result in maximum circuit current to flow.**
- **This effect is called the Zener Breakdown or Avalanche Breakdown**

Reverse Biasing

- **The I-V Characteristic curve of the P-N diode illustrates the current (I) and voltage relationship of the diode under forward and reverse bias conditions.**
- **It is obtained by sweeping the voltage across the diode from negative to positive whiles measuring the corresponding currents.**



Definition of Terms

❖ **Knee voltage or Cut-in Voltage**

It is the forward voltage at which the diode starts conducting.

❖ **Breakdown voltage**

It is the reverse voltage at which the diode (p-n junction) breaks down with sudden rise in reverse current.

❖ **Peak-inverse voltage (PIV)**

- **It is the maximum reverse voltage that can be applied to a p-n junction without causing damage to the junction.**
- **If the reverse voltage across the junction exceeds its peak-inverse voltage, then the junction gets destroyed because of excessive heat.**

Definition of Terms

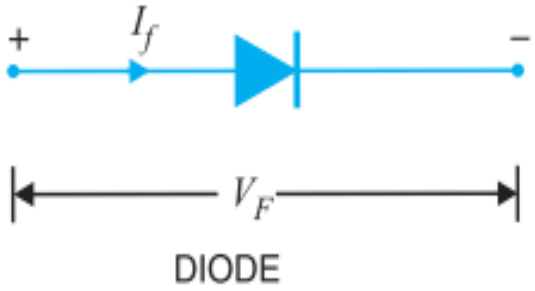
- In rectification, one thing to be kept in mind is that, care should be taken that reverse voltage across the diode during negative half cycle of A.C. does not exceed the peak-inverse voltage of the diode.
- It is usually safer to select a diode that has reverse breakdown voltage at least 50% greater than the expected PIV.

Maximum Forward current/Current handling Capacity

- It is the maximum instantaneous forward current that a p-n junction can conduct without damaging the junction.
- If the forward current is more than the specified rating, then the junction gets destroyed due to overheating.

Equivalent Circuit of a Diode

- When the forward voltage V_F is applied across a diode, it will not conduct till the potential barrier V_B at the junction is overcome.
- When the forward voltage exceeds the potential barrier voltage, the diode starts conducting as shown below.
- The forward current I_F flowing through the diode causes a voltage drop in its internal resistance r_f.



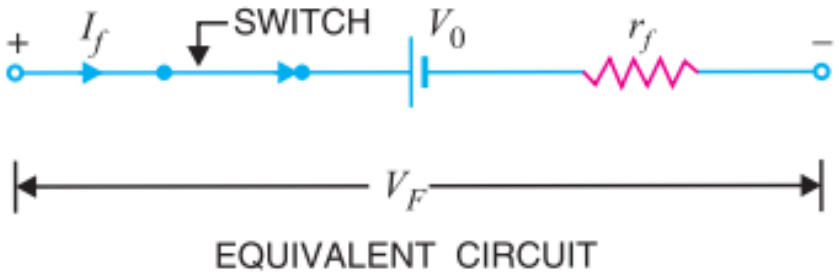
Equivalent Circuit of a Diode

- **Hence, the forward voltage V_F applied across the actual diode has to overcome**

  i. potential barrier VB

  ii. internal drop r_f

- **Hence, the approximate equivalent circuit for a diode is a switch in series with a battery V_B and internal resistance r_f as shown below. For most applications, the internal resistance r_f can be ignored.**



EQUIVALENT CIRCUIT

Equivalent Circuit of a Diode

- An ideal diode is one which behaves as a perfect conductor when forward biased and as a perfect insulator when reverse biased.

- This implies that the forward resistance $r_f = 0$ and potential barrier V_B is considered negligible.

Example 1

An AC voltage of peak value 20V is connected in series with a silicon diode and load resistance of 500 Ω.

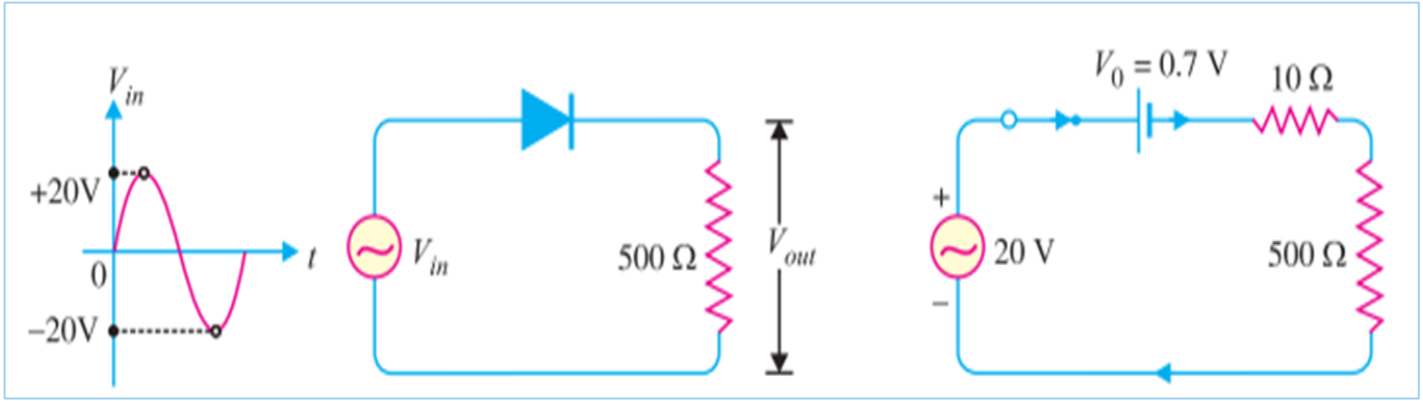
If the forward resistance of diode is 10 Ω, find under real and ideal conditions the:

(i) Peak current through diode

(ii) Peak output voltage

Solution

The diode will only conduct during the positive half-cycles of AC input voltage. The equivalent circuit is shown below:



Solution

Real Conditions

The peak current through the diode will occur at the instant when the input voltage reaches positive peak i.e. $V_{in} = V_F = 20V$

i. Peak Current

$$V_F = V_B + I_{fpeak}(r_f + R_L)$$

$$I_{fpeak} = \frac{V_F - VB}{r_f + RL} = \frac{20 - 0.7}{10 + 500} = 37.8mA$$

ii. Peak output voltage

$$V_{Pout} = I_{fpeak} \times R_L = 37.8mA \times 500\Omega = 18.9V$$

Solution

Ideal Conditions

i. Peak Current

For an ideal diode, put $V_B = 0$ and $r_f = 0$

$$V_F = I_{fpeak} \times R_L$$

$$I_{fpeak} = \frac{V_F}{R_L} = \frac{20}{500} = 40mA$$

ii. Peak output voltage

Peak Output Voltage $= I_{fpeak} \times R_L = 20V$

Thanks!

Any questions?